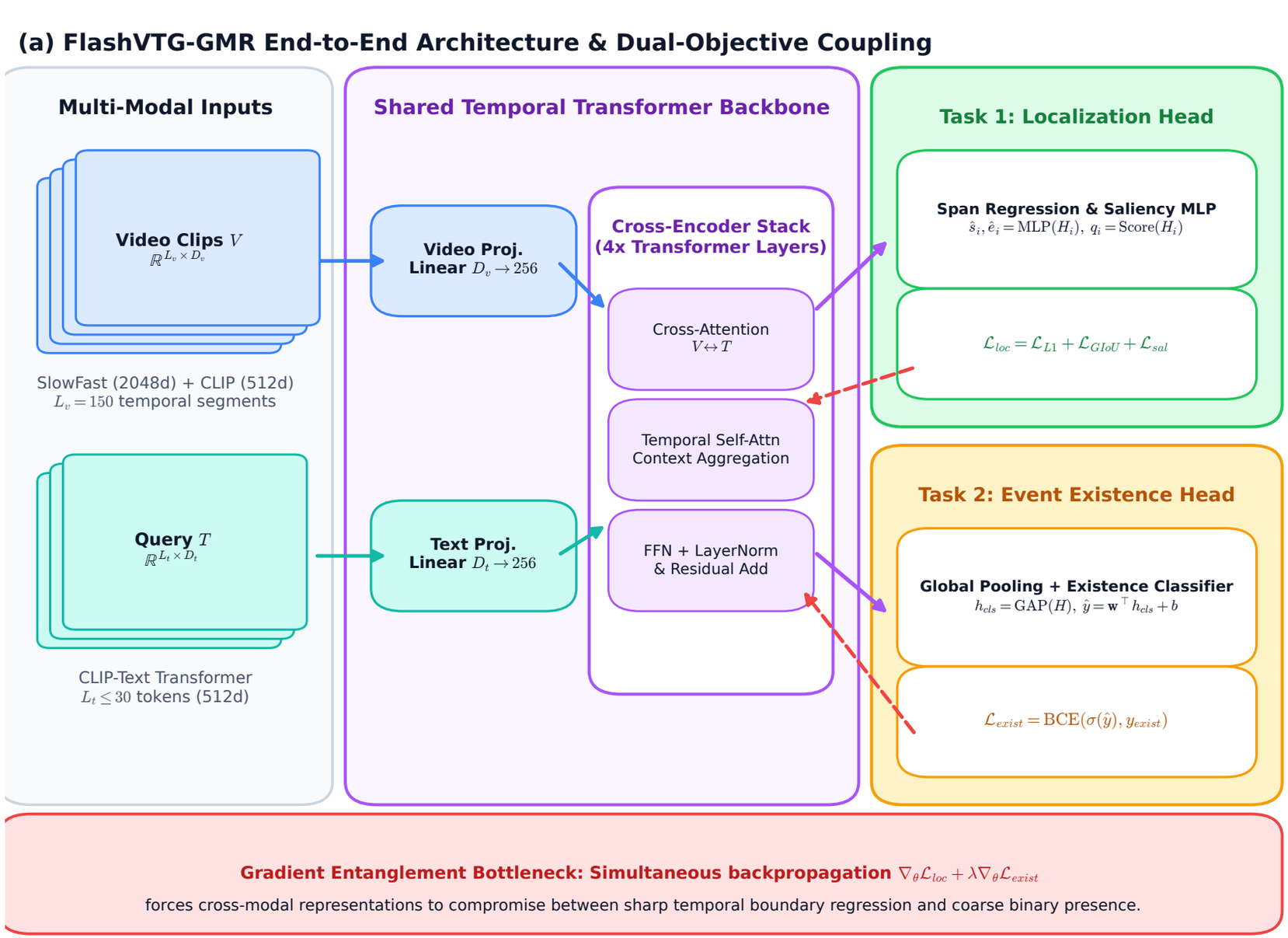
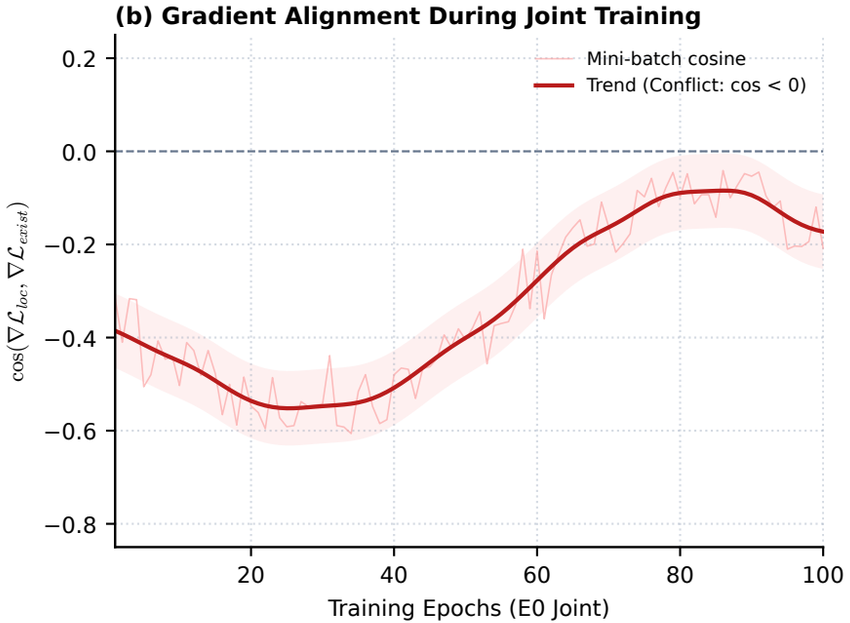
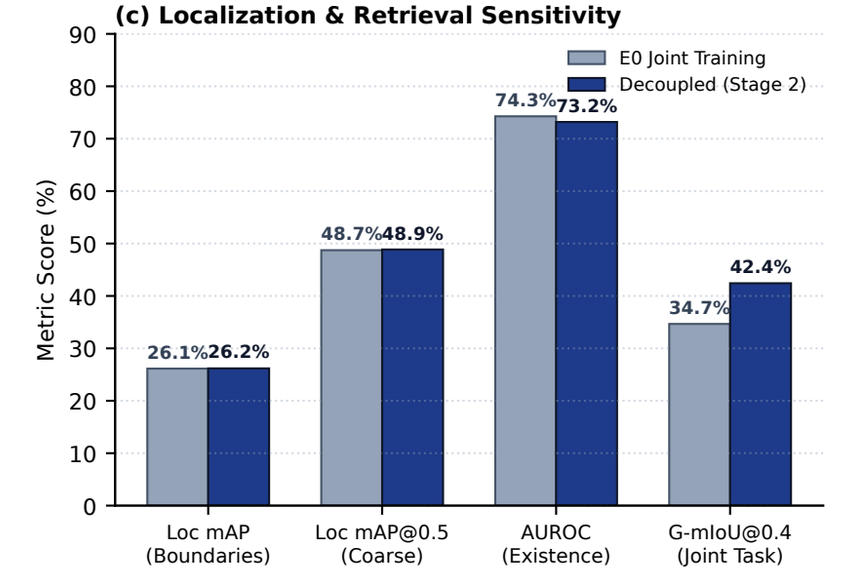

(a) FlashVTG-GMR End-to-End Architecture & Dual-Objective Coupling
Multi-Modal Inputs
Video Clips V
$\mathbb{R}^{L_v \times D_v}$
SlowFast (2048d) + CLIP (512d)
$L_v = 150$ temporal segments
Query T
$\mathbb{R}^{L_t \times D_t}$
CLIP-Text Transformer
$L_t \leq 30$ tokens (512d)
Shared Temporal Transformer Backbone
Video Proj.
Linear $D_v \to 256$
Text Proj.
Linear $D_t \to 256$
Cross-Encoder Stack
(4x Transformer Layers)
Cross-Attention
$V \leftrightarrow T$
Temporal Self-Attn
Context Aggregation
FFN + LayerNorm
& Residual Add
Task 1: Localization Head
Span Regression & Saliency MLP
$\hat{s}_i, \hat{e}_i = \mathrm{MLP}(H_i),\ q_i = \mathrm{Score}(H_i)$
$\mathcal{L}_{loc} = \mathcal{L}_{L1} + \mathcal{L}_{GIoU} + \mathcal{L}_{sal}$
Task 2: Event Existence Head
Global Pooling + Existence Classifier
$h_{cls} = \mathrm{GAP}(H),\ \hat{y} = \mathbf{w}^\top h_{cls} + b$
$\mathcal{L}_{exist} = \mathrm{BCE}(\sigma(\hat{y}), y_{exist})$
Gradient Entanglement Bottleneck: Simultaneous backpropagation $\nabla_\theta \mathcal{L}_{loc} + \lambda \nabla_\theta \mathcal{L}_{exist}$
forces cross-modal representations to compromise between sharp temporal boundary regression and coarse binary presence.


(b) Gradient Alignment During Joint Training
Mini-batch cosine
Trend (Conflict: cos < 0)
$\cos(\nabla \mathcal{L}_{loc}, \nabla \mathcal{L}_{exist})$
0.2
0.0
−0.2
−0.4
−0.6
−0.8
20
40
60
80
100
Training Epochs (E0 Joint)


(c) Localization & Retrieval Sensitivity
E0 Joint Training
Decoupled (Stage 2)
Metric Score (%)
90
80
70
60
50
40
30
20
10
0
26.1%
26.2%
48.7%
48.9%
74.3%
73.2%
34.7%
42.4%
Loc mAP
(Boundaries)
Loc mAP@0.5
(Coarse)
AUROC
(Existence)
G-mIoU@0.4
(Joint Task)